

Specifications and Features of Version 2.17

Sample Rate	Variable, single shot 1 MHz max, interleaved 417 kHz max, time equivalent 5 MHz max
Bandwidth	5 MHz analogue bandwidth, 500 kHz practical bandwidth
Number of Channels	Two
Sample Depth	256 bytes per channel, 128 bytes per channel when interleaved
Sample Modes	Single shot, interleaved channel, and time equivalent
Calibration	Voltage offset
Configuration Settings	Configuration of default and saving of settings for future use
Hardware Connection	Any parallel, serial or USB port (using serial over USB)
Coupling	AC and DC, reflected on PC
Channel gain	Gain of 1, 2, and 5, reflected on PC
Trigger	Variable level, slope and timing
Frequency Spectrum	Yes
Waveform Reconstruction	Spline, triangle, square, point and sinc
XY view	Yes
Channel offset and volt per division settings	Yes
Cursors	Yes
Math functions	Addition, subtraction, auto period, auto peak-to-peak, etc.
Waveform export	Comma delimited
Waveform logging	Comma delimited logfile containing waveform and measurement information

Latest features in Version 2.17

1. Added support for features in Paul Messer's DSOScope Rev 3 implementation:

(a) Pretrigger in XY sampling mode (chop sampling) available at around 500 μ s per division

(b) An interrupt-driven sampling mode available at around 500 μ s per division capable of drawing the channel as it is captured

(c) Extended range of low-end sample rates

(d) Integration of attenuation and trigger source switch status into the software panel

2. Improvements to the protection circuitry on the analogue front-end improve protection and the analogue bandwidth. In October, a user pointed out that the oscilloscope was bandwidth-limited. The bandwidth limit was around 1 MHz.

However, you could see the bandwidth limitation readily with a lower-frequency square wave when the square wave is sampled at 5MHz time-equivalent sampling mode. The changes are simple and can be implemented without re-etching boards, but by changing out four components. The changes raised the analogue bandwidth of the scope to 5 MHz or higher.

3. Fixed a scaling bug to the spectrum window introduced in the 2.16 release

4. Changed trigger indication

5. Serial and USB communications improvements. The 2.16 release resulted in slower waveform refresh rates because of a bug that caused every waveform update to utilise a timeout. In short, the timeout was killing the refresh rate. Now the timeout is avoided.

6. Window mouseover and cursor measurements now indicate the time in reference to the trigger instead of the left hand of the display

7. Corrected a bug in hardware testing

8. Corrected a bug that could misreport the first sample in XY mode at higher sample rates

9. Changed the functionality of the coarse time per division and volts per division adjustments to jump according to a traditional 1, 2, 5 sequence

10. Additional cubic spline reconstruction method. This method provides a smoother reconstruction than straight line reconstruction while introducing less oscillation in the reconstruction than the sinc reconstruction method.

11. Addition of a high-resolution mode. In many cases, this effectively adds around 1.5 bits of resolution to the scope. While the scope uses an 8-bit ADC, the high-resolution mode uses averaging to achieve nearly a 10-bit resolution.

Working

The scope connects with your Windows computer and communicates with the OS to display the captured waveform in both time and fre-

quency domains. It combines a high-frequency analogue stage along with an AC/DC coupling and three gains, a MAX118 digital-to-analogue converter, a trigger circuit, a PIC16F877A MCU and some PC interfacing hard-

ware. Latest board layouts and data-sheets of the various components are available at: <http://jonw0224.tripod.com/ppmscope.html> •

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